CALCULATING FAMILY EXPENSES USING SERVICENOW

Ideation Phase

The ideation phase focused on identifying a practical, real-life scenario that demonstrates ServiceNow's capability to automate and manage data efficiently. After analyzing several ideas, the team decided to build a simple yet useful Family Expense Management System using ServiceNow. The goal was to design an application that helps families or individuals record, track, and analyze their daily spending patterns. By using ServiceNow tables, relationships, and business rules, this project aims to automate the process of calculating and categorizing family expenses dynamically.

Key Ideas:

- Create a central table to record overall family expenses.
- Design a secondary table to store daily expenses for each member.
- Build a relationship between both tables for data tracking.
- Use business rules to calculate total expenses automatically.
- Provide transparency and accuracy through automated updates.

Problem Statement

Managing daily and monthly family expenses manually often leads to confusion, calculation errors, and lack of visibility into total spending. There is a need for an automated system that records expenses, categorizes them, and calculates the total dynamically. This project leverages ServiceNow's automation, relationships, and business rule capabilities to simplify family expense tracking.

Objectives

• To design and implement a ServiceNow-based Family Expense Management application.
• To create two custom tables — Family Expenses and Daily Expenses.
• To establish a relationship between the two tables for linked data representation.
• To configure business rules that calculate total family expenses automatically.
• To demonstrate the use of update sets, relationships, and UI configurations in ServiceNow.

Expected Outcomes

• A dynamic ServiceNow application that records and calculates family expenses.
• Automated total calculation through business rules.
• A clear and structured form view for entering and managing expense data.
• Improved data accuracy and accessibility.
• Understanding of table relationships and automation concepts in ServiceNow.

Scope of the Project

The scope is limited to building and testing a small-scale ServiceNow application that calculates total family expenses based on related daily expense records. This project demonstrates database design, UI configuration, and automation in ServiceNow.

Project Planning Phase

Planning Activities:
1. Setting up developer instance and creating a new application scope.

2. Creating an update set to capture all changes.
3. Designing tables and relationships.
4. Configuring UI forms and related lists.
5. Writing business rules for automated calculations.
6. Testing and validating results.

Timeline and Execution Plan

Activity	Duration	Responsible
Instance Setup	**1 Day**	**Admin**
Update Set Creation	**1 Day**	**Admin**
Table Creation	**2 Days**	**Developer**
Relationship Configuration	**1 Day**	**Developer**
Business Rule Configuration	**1 Day**	**Developer**
Testing & Documentation	**2 Days**	**Admin**

Implementation Phase

Step 1: **Setting Up Instance**

A ServiceNow developer instance was requested and activated from developer.servicenow.com. After logging in, the instance was personalized with the necessary modules and permissions.

Step 2: **Creation of New Update Set**

To capture configuration changes:
1. Navigate to System Update Sets → Local Update Sets → New.
2. Name: Family Expense Tracking.
3. Set as Current Update Set. All further changes were saved here for migration and version control.

Step 3: **Creation of Table – Family Expenses**

1. Navigate to System Definition → Tables → New.
2. Table Name: Family Expenses.
3. Key Fields:
   - Number (Auto Number, Read-Only)
   - Family Member Name (Reference to sys_user)
   - Date (Mandatory)
   - Amount (Mandatory)
   - Category (Choice field)
   - Description (String)

Step 4: **Creation of Table – Daily Expenses**

1. Navigate to System Definition → Tables → New.
2. Table Name: Daily Expenses.
3. Key Fields:
   - Number (Auto Number, Read-Only)
   - Date (Mandatory)
   - Expense Type (Choice field)
   - Amount (Mandatory)
   - Related Family (Reference to Family Expenses)

Step 5: **Creation of Relationship**

1. Navigate to System Definition → Relationships → New.
2. Name: Daily Expenses.
3. Applies to Table: Family Expenses.
4. Related Table: Daily Expenses.
5. Save to establish the one-to-many relationship.

Step 6: **Configuring Related List on Family Expenses**

1. Open any record from Family Expenses.
2. Right-click the header → Configure → Related Lists.
3. Add Daily Expenses to the selected related lists.
4. Save and verify that related Daily Expenses now appear under each

Family Expense record.

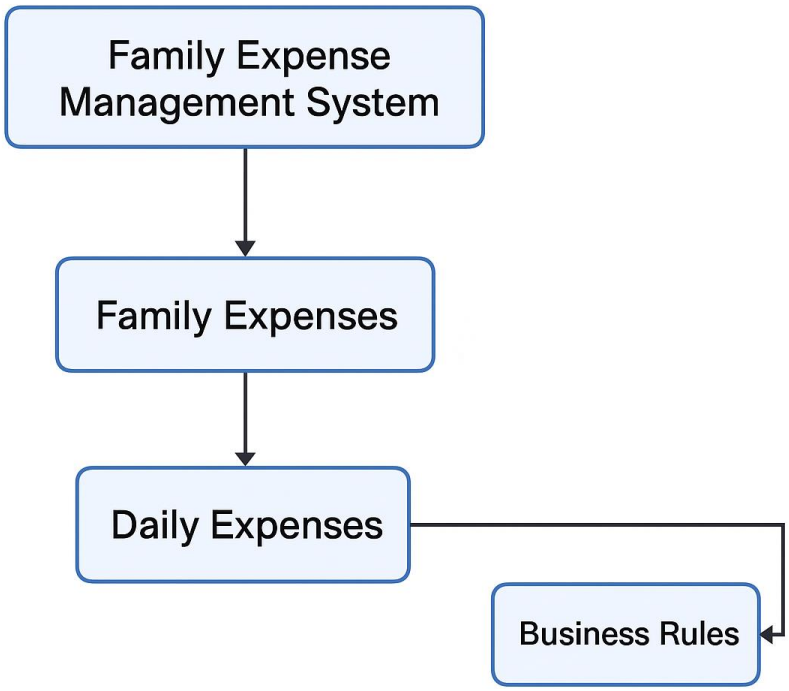
Step 7: **Creation of Business Rules**

1. Navigate to System Definition → Business Rules → New.
2. Name: Calculate Total Expense.
3. Table: Family Expenses.
4. When: Before Update/Insert.
5. Script: Aggregates total from related Daily Expenses records and updates the Total field automatically.
6. Save and test.

Step 8: **Configuring Relationship Validation**

After setting up the business rules, verify that when a Daily Expense record is created or updated, the related Family Expense record automatically recalculates the total amount.

Architecture

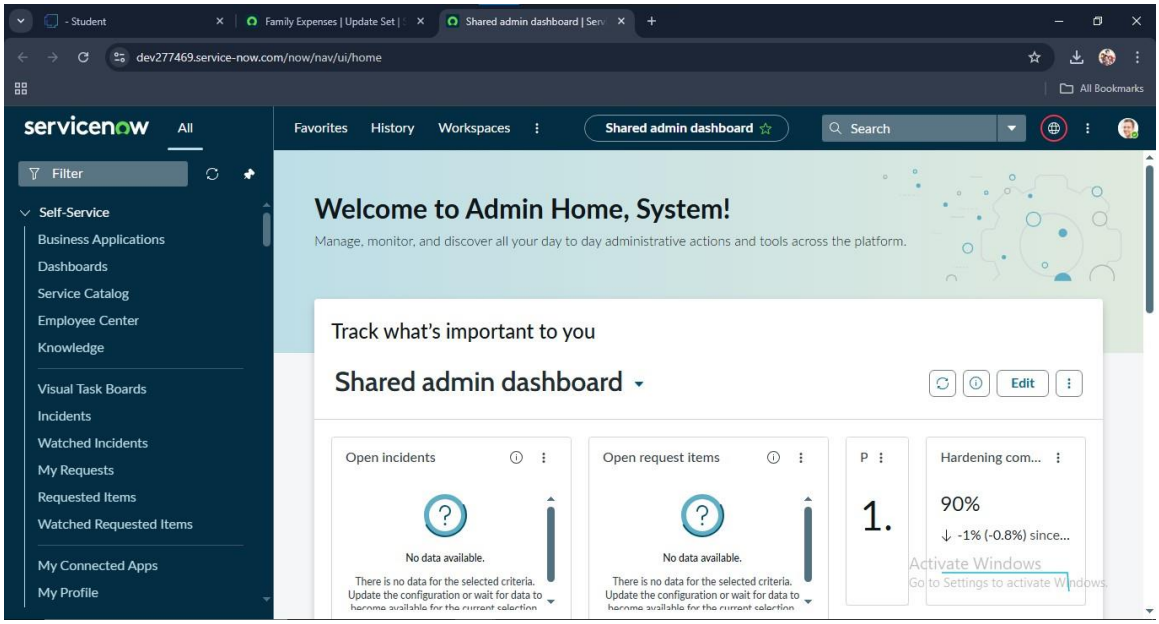


Conclusion

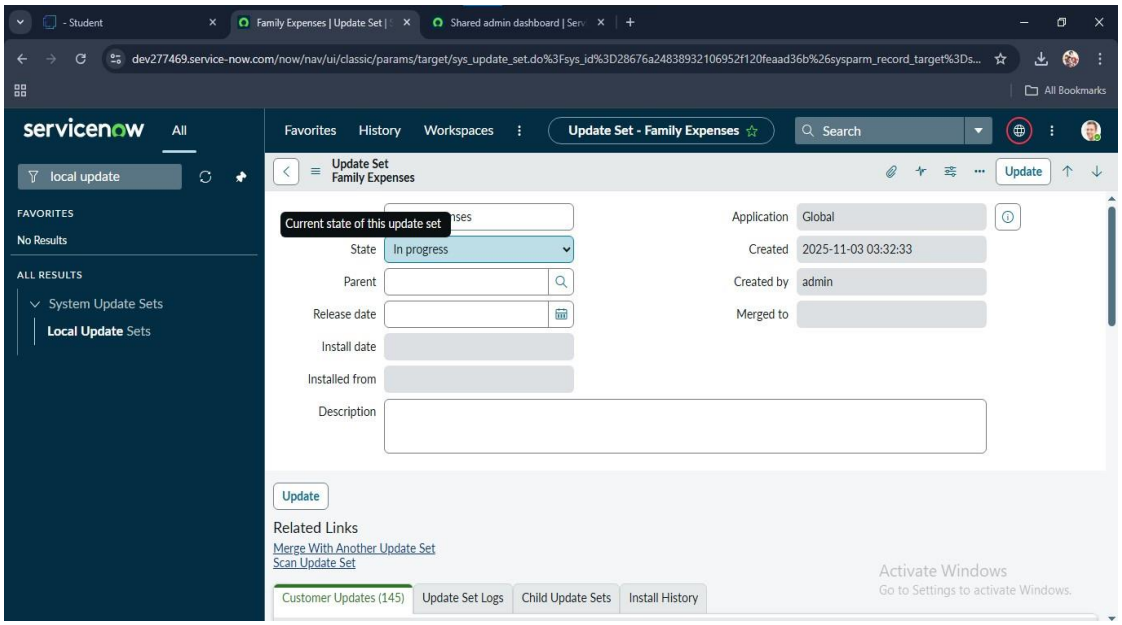
The project “Calculating Family Expenses Using ServiceNow” demonstrates how ServiceNow can be used beyond IT workflows to manage daily life automation. Through custom tables, relationships, and business rules, the system efficiently calculates and updates family expenses in real-time. This project highlights the power of ServiceNow as a low-code platform capable of solving real-world automation problems through structured design and workflow configuration.

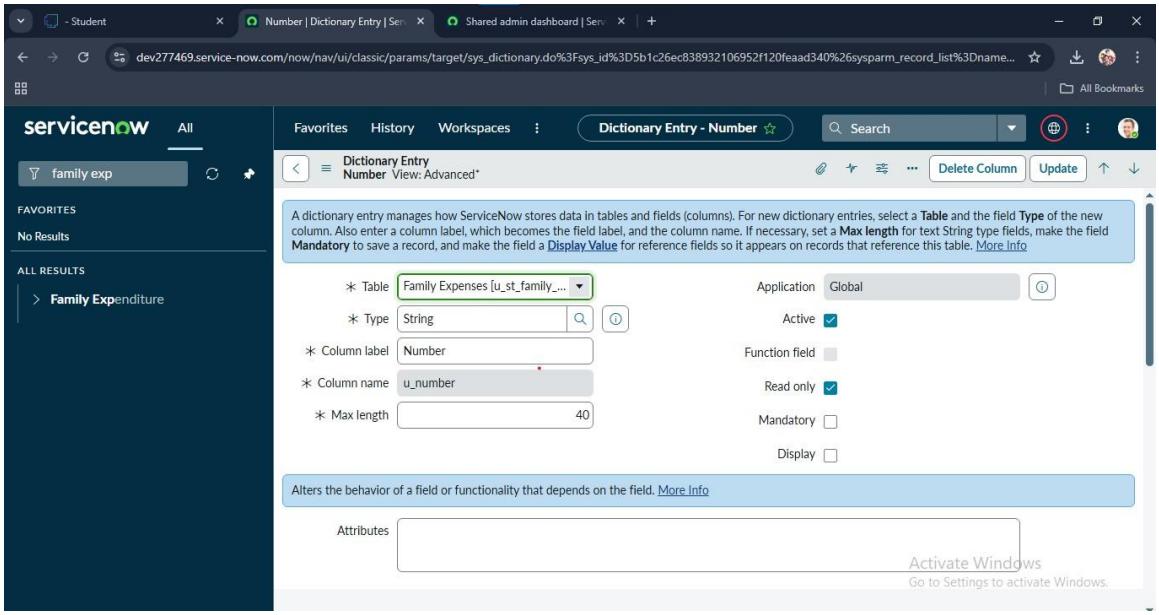
Evidence

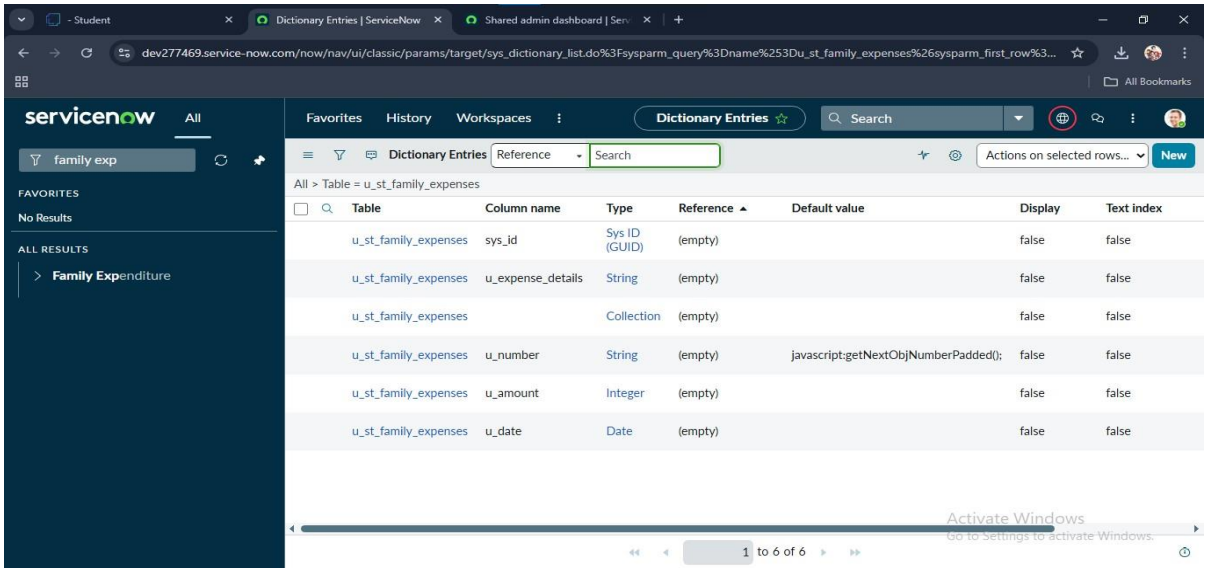
- Setting up ServiceNow Instance

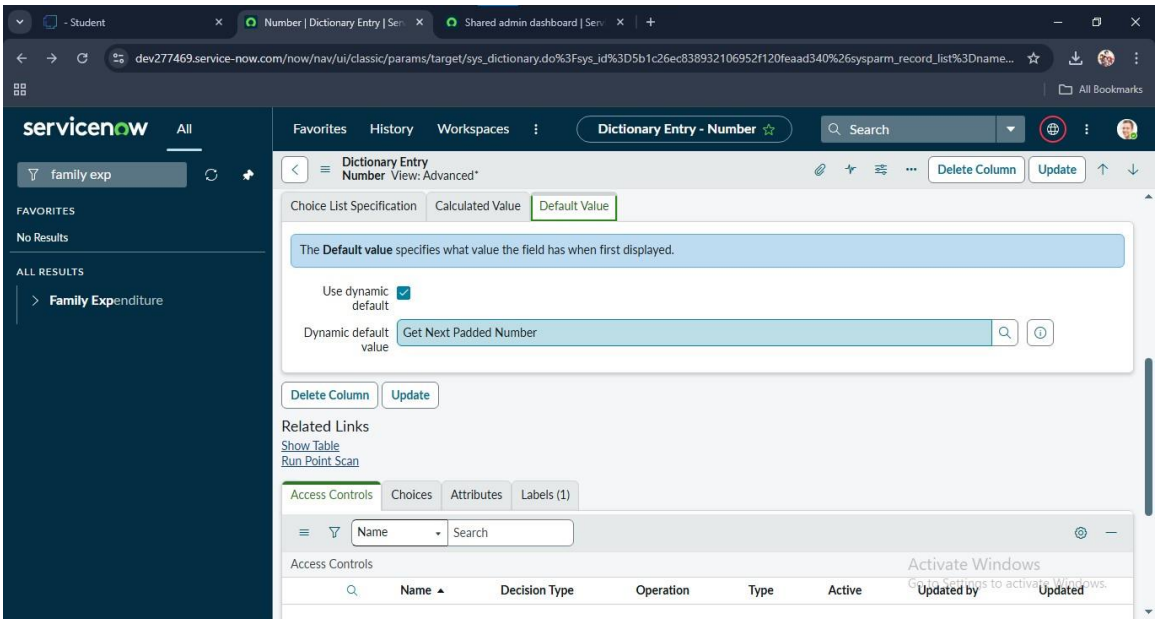


- Creation of New Update Set

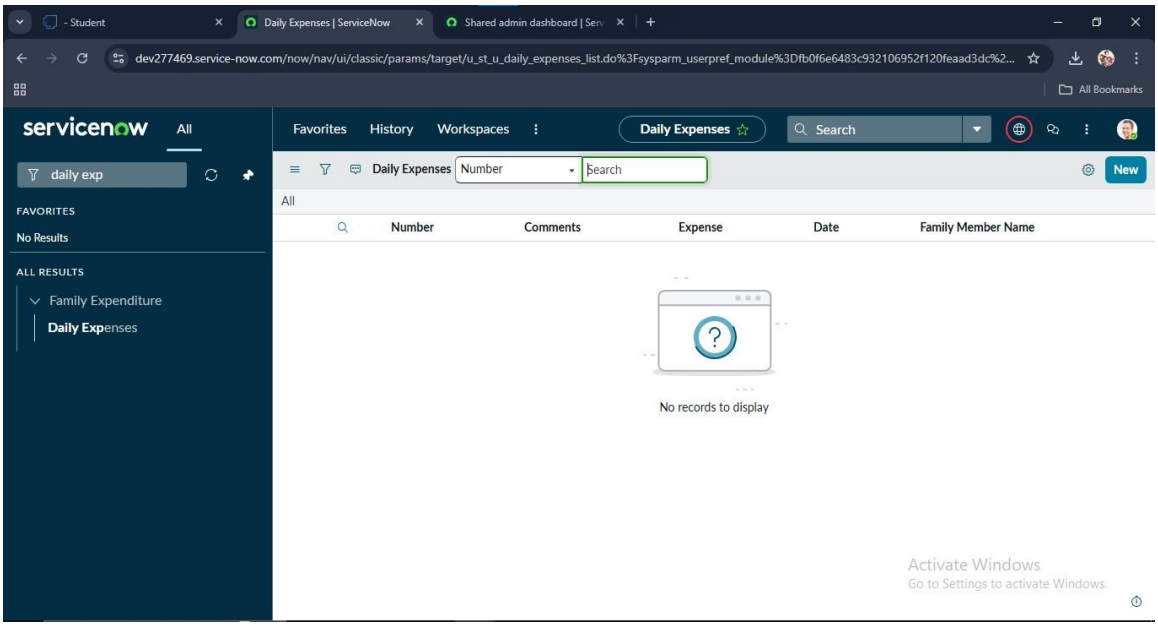




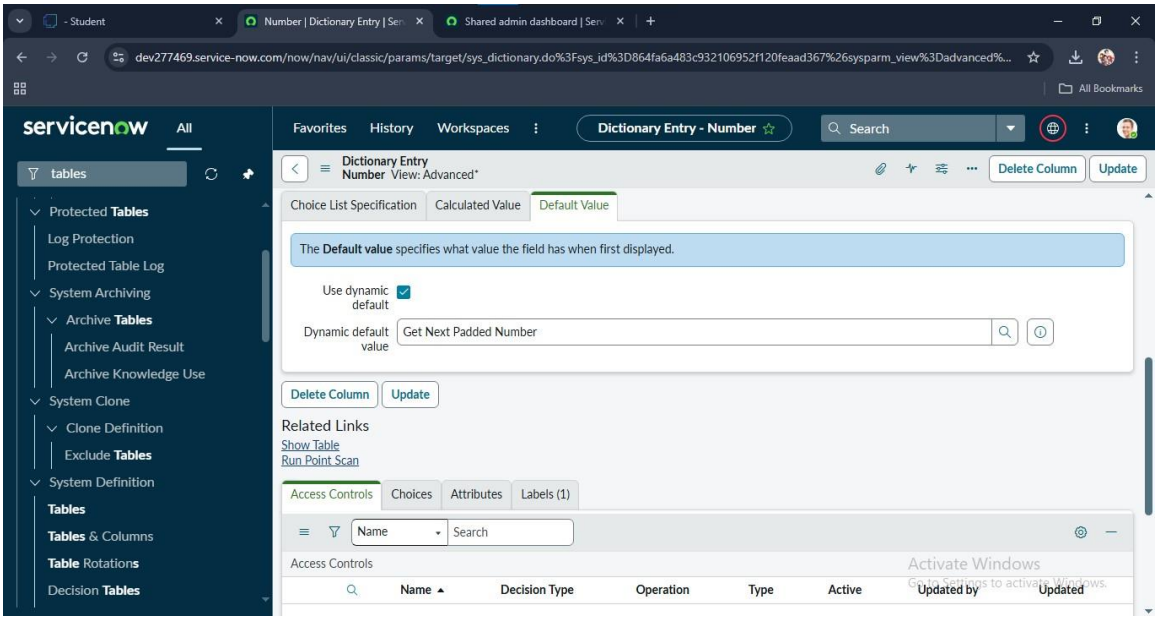


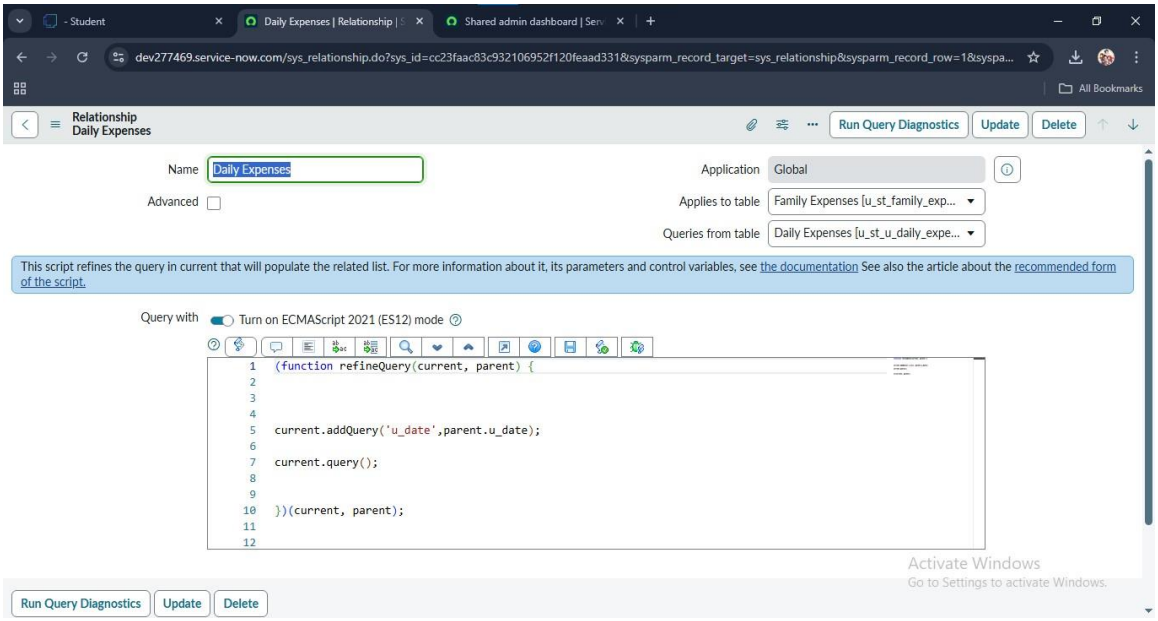


- Creation of Table(Daily Expenses)

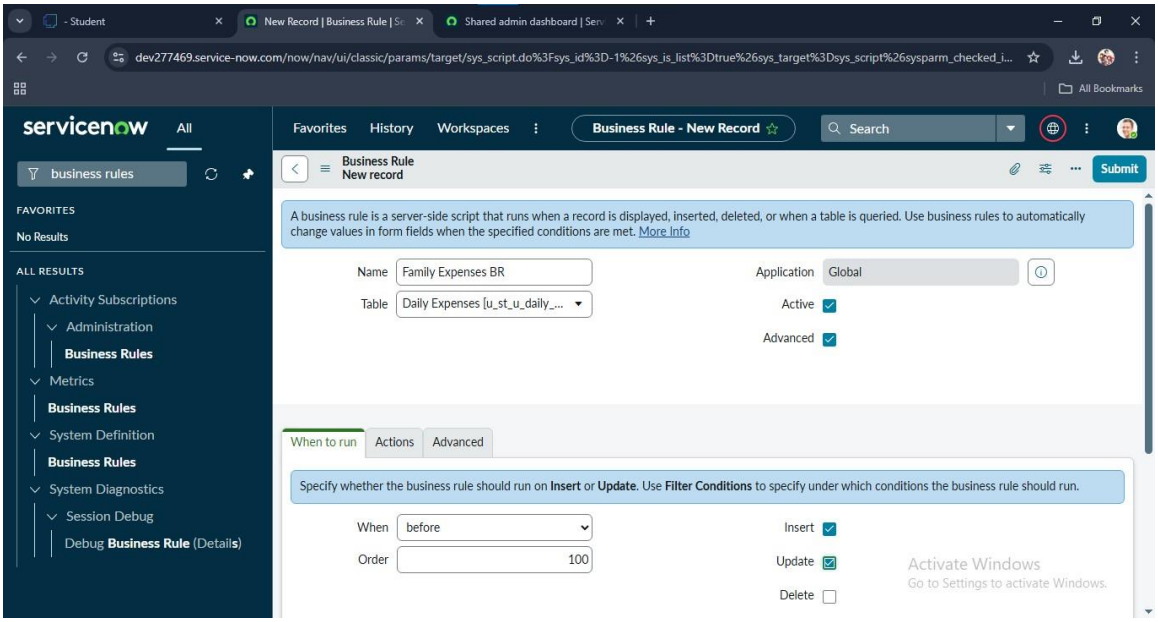


- Creation of Relationship





- Creation of Business Rules



Conclusion

The project "Calculating Family Expenses Using ServiceNow" successfully demonstrates how ServiceNow can be utilized beyond traditional IT service management — as a powerful platform for process automation and data handling.By designing two interconnected tables (Family Expenses and Daily Expenses), configuring relationships, and

implementing business rules, the system automatically calculates total family expenses based on individual daily records.Through this project, the following outcomes were achieved:

- Seamless linkage between parent and child records using relationships.
- Real-time calculation and data updates via business rules.
- Improved visibility and organization of expense data.
- Hands-on understanding of ServiceNow components such as tables, update sets, and form configuration.

Overall, this project highlights ServiceNow's versatility as a low-code platform capable of solving everyday data management problems efficiently. The Family Expense Management application provides a practical example of how automation can simplify record maintenance, ensure accuracy, and enhance productivity.

Submitted by,

- S.SHYAM SHANKAR DEPT: BTECH (AI&DS) - 961222243022
- SIVA KUMAR DEPT: BTECH (AI&DS) - 961222243023
- P.DHANUSH DEPT: BTECH (AI&DS) - 961222243006